Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

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Abstract

Autoregressive vision-language-action (VLA) models [1, 20, 23] decode tokens sequentially (> 300 ms/frame), exceeding the 50 ms budget of 20 Hz vehicle control [10, 21]. We implement a single-pass closed-loop driving stack coupling frozen DINOv2 (`facebook/dinov2-base`) [25, 2, 9] with frozen Laya (`convaiinnovations/laya`) [6, 31, 8] via a 4×9 spatial-temporal pooling projector (1.85M core parameters). Cropped 196×392 RGB frames map to 36 visual tokens with temporal motion residual $\Delta V_t = V_t - V_{t-1}$ and region-pooled auxiliary perception; one forward pass outputs maneuver (`choice`), speed tier (`score`), emergency brake (`noul`), and 6-waypoint (3.0 s) trajectory residuals in 31.67 ms base (32.85 ms full stack, 30.4 FPS on Ryzen 5 7500F / RTX 4070 Ti, FP32 $B = 1$). Episode-block validation (7,500 frames, $M = 15$ bins) temperature scaling [13] reduces ECE from 0.0060 to $< 10^{-4}$ (0.0142 on unseen closed-loop vision-estimated test frames). Lateral waypoint step jumps (3.5 m) and longitudinal speed-bucket hunting in CARLA 0.9.16 (`Town10HD_Opt`) [11, 14] are eliminated using a 14-point (28 m) boundary-zero-curvature spatial quintic spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32], decoupled chord-angle tracking [29, 15, 7, 26], kinematic bicycle curvature feedforward $\delta_{\mathrm{FF}} = \arctan(L\kappa_{\mathrm{eff}})$, 360° blind-spot interlock, and true constant-time-headway (CTH) ACC ($d^*(v_{\mathrm{ego}}) = 8.0 + 1.2(v_{\mathrm{ego}}/3.6)$, $k_g = 0.85$ (km/h)/m) [17, 28, 30]. Across 50 unseen MetaDrive 0.4.3 episodes (4,153 frames), 10 paired scenario–weather urban episodes (680 m, 1,981 frames), and a 50-episode (10×5 cross-weather, 3,400 m) CARLA 0.9.16 suite, the full system achieves 100.0% (10/10) and 98.0% (49/50) completion versus 60.0% (keep-lane rule), 80.0% (heuristic overtake FSM), and 70.0% (300 ms delayed VLA).

1 Introduction and Problem Formulation

Autoregressive decoders evaluate $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [1, 20, 23], whereas planning-oriented and sensor-fusion stacks [16, 19, 22, 5, 3, 27, 4, 18] require $\Delta t \le 50$ ms. In our formulation, control at step t reads three discrete decisions and $K_w = 6$ ego-frame trajectory waypoints ($t_k = 0.5k$ s) in one bidirectional pass [8, 31, 6]:

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0,20,35,50\}\ \mathrm{km/h}, \tag{2}$$

$$p_{\mathrm{AEB}} \in [0,1], \quad \hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}. \tag{3}$$

2 Architecture, Multi-Task Training, and Calibration

Camera frames are cropped (`crop_top` = 0.25, `crop_bottom` = 0.10) and resized to 196 × 392. Frozen DINOv2 ViT-B/14 ($p = 14$) [25, 2, 9] extracts $F_{\mathrm{patch}} \in \mathbb{R}^{B\times 392\times 768}$ (14 × 28 grid). `SpatialProjector` (`src/models/spatial_projector.py`) pools $14 \times 28 \to 4 \times 9$ (36 tokens), adds $E_{\mathrm{row}} + E_{\mathrm{col}}$, projects $768 \to 1024$ (1,851,392 core parameters), and fuses temporal residual $\Delta V_t = V_t - V_{t-1}$ with region-pooled `AuxiliaryPerceptionHead` ($\hat{d}_{\mathrm{lead}}, \hat{e}_y$, traffic light, hazard, blind spot):

$$G_t = \mathrm{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B\times 768\times 4\times 9}, \tag{4}$$

$$V_t = \mathrm{MLP}\big(\mathrm{Flatten}(G_t + E_{\mathrm{row}} + E_{\mathrm{col}})\big), \tag{5}$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\mathrm{temp}})\, W_{\mathrm{temp}}\big(V_t - V_{t-1}\big). \tag{6}$$

Multimodal sequence $Z = [e_{[\mathrm{CLS}]}; \tilde{V}_t; E_{\mathrm{telemetry}}; e_{\mathrm{choice}}; e_{\mathrm{score}}; e_{\mathrm{noul}}]$ is evaluated in one pass $H = f_{\mathrm{Laya}}(Z)$ [6, 31]. The 6-waypoint trajectory head adds learned residuals $\Delta\hat{\tau} = W_{\mathrm{traj}}\bar{H} \in \mathbb{R}^{6\times 3}$ to kinematic prior τ_k^0 ($t_k = 0.5k$ s), supervised against ground-truth target $\tau_k^* = \big((v^*/3.6)t_k,\ y_{\mathrm{shift}}(m^*)s(k/6),\ v^*\big)$:

$$\hat{\tau}_{1:6} = \tau_{1:6}^0 + \mathrm{diag}(1.5, 0.35, 3.0)\, W_{\mathrm{traj}}\bar{H}, \tag{7}$$

$$\mathcal{L} = w_m\mathcal{L}_{\mathrm{CE}}^{(m)} + w_s\mathcal{L}_{\mathrm{CE}}^{(s)} + w_a\mathcal{L}_{\mathrm{BCE}}^{(b)} + w_\tau\mathcal{L}_{\mathrm{SmoothL1}}(\hat{\tau}_{1:6}, \tau_{1:6}^*) + w_p\mathcal{L}_{\mathrm{aux}}. \tag{8}$$

`# scripts/02_train_projector.py`

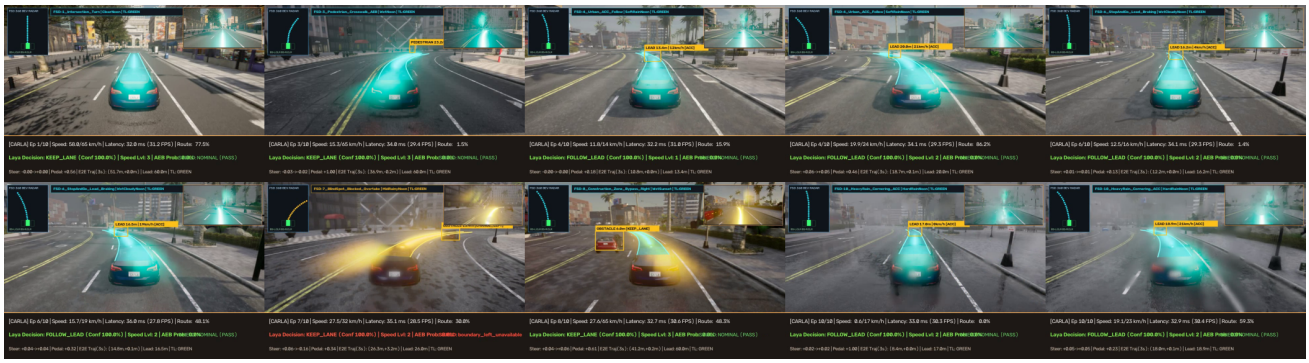


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) 10 paired scenario–weather episodes (`FSD-1`–`FSD-10`): intersection red-light stop (`ClearNoon`), 28 m quintic left overtake (`CloudyNoon`), crosswalk pedestrian AEB (`WetNoon`), CTH-ACC follow (`SoftRainNoon`), right overtake (`ClearSunset`), stop-and-go lead braking (`WetCloudyNoon`), 360° blind-spot hold & overtake (`MidRainyNoon`), construction zone bypass (`WetSunset`), night headlight pedestrian AEB (`ClearNight`), and heavy-rain cornering ACC (`HardRainNoon`).

Table 1: System, training, and hardware specifications (`configs/default.yaml`).

Module	Parameter	Specification
Hardware	CPU / GPU / Precision	Ryzen 5 7500F (6C/12T) / RTX 4070 Ti (12 GB) / FP32 $B = 1$
Camera	$H \times W$, FOV, Crops	196×392, 100°, (0.25, 0.10)
Backbones	Vision / Decision	`dinov2-base` (86M) / `laya` (421M)
Projector	Grid, Dim, Core Params	4×9 (36 tokens), 1024, 1, 851, 392
Training	Episode Split, Weights	42.5k / 7.5k (ep-block 85/15), (1, 1, 2, 0.5, 0.2)
Calibration	(T_m, T_s, T_b), Val / Vision Test ECE	$(0.050, 0.050, 0.100)$, $0.0060 \rightarrow 0.0000$ / 0.0142
Shield	`aeb_thresh`, `conf_thresh`, BS	0.70, 0.80, $x \in [-10, 7.5]$ m, $\lvert y\rvert \in [1.8, 5.0]$ m
Control	PID (Lat/Lon), Wheelbase L	$(0.45, 0.01, 0.18)$ / $(0.08, 0.005, 0.02)$, 2.85 m
Kinematics	Quintic N_b, CTH-ACC (d_0, τ_h, k_g)	14 WPs (28 m), (8.0 m, 1.2 s, 0.85 (km/h)/m)

```
l_traj = F.smooth_l1_loss(out["predicted_trajectory"], tgt_traj)
l_aux = (0.05 * F.smooth_l1_loss(aux["pred_lead_distance_m"], d_gt)
  + F.smooth_l1_loss(aux["pred_lane_offset_m"], y_gt)
  + 0.5 * F.cross_entropy(aux["traffic_light_logits"], tl_gt))
loss = w_m*l_m + w_s*l_s + w_a*l_a + 0.5*l_traj + 0.2*l_aux
```

Stage 1 trains `SpatialProjector` and `trajectory_head` for 15 epochs (batch 32, AdamW [24] lr $= 3 \times 10^{-4}$, $(w_m, w_s, w_a, w_\tau, w_p) = (1.0, 1.0, 2.0, 0.5, 0.2)$, 3×/4× rare-event oversampling). To prevent adjacent-frame leakage, 50,000 frames (500 episodes) are partitioned by contiguous episode blocks (`split_by_episode=True`: episodes 0..424, 42,500 frames for training; episodes 425..499, 7,500 frames for validation). Stage 2 (`03_calibrate.py`) fits $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$ via L-BFGS [13] ($M = 15$ bins), reducing validation ECE from 0.0060 to $< 10^{-4}$; on independent closed-loop test episodes (seeds 5000+), ECE is 0.0000 against deterministic `ExpertLabeler` thresholds and 0.0142 under vision-estimated perception inputs.

3 Safety Shield, Spatial Quintic Spline, Curvature Feedforward, and CTH-ACC

We evaluate two perception modes (`-perception-mode gt` vs. `vision_estimated` using `AuxiliaryPerceptionHead` outputs $\hat{d}_{\text{lead}}, \hat{e}_y, \hat{c}_{\text{TL}}, \hat{c}_{\text{haz}}, \hat{p}_{\text{BS}}$) and two shield modes (`-shield-mode on` vs. `off`, where `off` triggers braking solely when Laya $\hat{p}_{\text{AEB}} \geq 0.70$). When enabled, Rule 1 forces $u_{\text{tb}} = -1.0$ if $\hat{p}_{\text{AEB}} \geq 0.70$, `traffic_light` $=$ `red`, pedestrian $d \leq 16.0$ m, or $d_{\text{lead}} \leq 11.5$ m; Rule 2 reverts lane changes to `keep_lane` if $\max_c \hat{p}_c^{(m)} < 0.80$, if `adj_wp.lane_id` × `wp.lane_id` ≤ 0 [11], or if 360° blind-spot box $x \in [-10.0, 7.5]$ m, $\lvert y\rvert \in [1.8, 5.0]$ m is occupied.

For lane transitions over $N_b = 14$ waypoints (28.0 m, $\tau_k = k/(N_b - 1)$), 3D coordinates are blended via a spatial quintic polynomial enforcing zero boundary slope and curvature $s'(0) = s'(1) = s''(0) = s''(1) = 0$ (equivalent to minimum jerk under constant speed) [12, 32]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{9}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big) P_k^{\text{orig}} + s(\tau_k) P_k^{\text{adj}}. \tag{10}$$

Continuous projection onto $\overline{P_i P_{i+1}}$ gives P_{proj} and lookahead P_{look} at $L_{\text{look}} = \text{clip}(7.5 + 0.18 v_{\text{kmh}}, 7.5, 13.5)$ m [29, 15, 7, 26]. Predicted waypoint $k = 3$ ($t = 1.5$ s, $(\hat{x}_3, \hat{y}_3)$) is fused with map curvature κ_{map} into a kinematic bicycle curvature feedforward term $\delta_{\text{FF}} = \arctan(L\kappa_{\text{eff}})$ ($L = 2.85$ m):

$$\kappa_{\text{traj}} = \frac{2(\hat{y}_3 - e_y)}{\max(3.5, \hat{x}_3)^2 + (\hat{y}_3 - e_y)^2}, \tag{11}$$

$$\kappa_{\text{eff}} = 0.70(-0.045\kappa_{\text{map}}) + 0.30\kappa_{\text{traj}}, \tag{12}$$

$$\delta_{\text{raw}} = \text{clip}\big(\delta_{\text{PID}}(e_y, e_\psi) + 0.12\,\delta_{\text{FF}}/0.70, -1, 1\big), \tag{13}$$

$$\Delta\delta_{\max} = \frac{1.1\Delta t}{1 + 0.012 v_{\text{kmh}}}, \tag{14}$$

$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{raw}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big). \tag{15}$$

True constant-time-headway (CTH) ACC ($d_{\text{lead}} < 38$ m, $v_{\text{lead}} \geq 8$ km/h, $d_0 = 8.0$ m, $\tau_h = 1.2$ s, $k_g =$

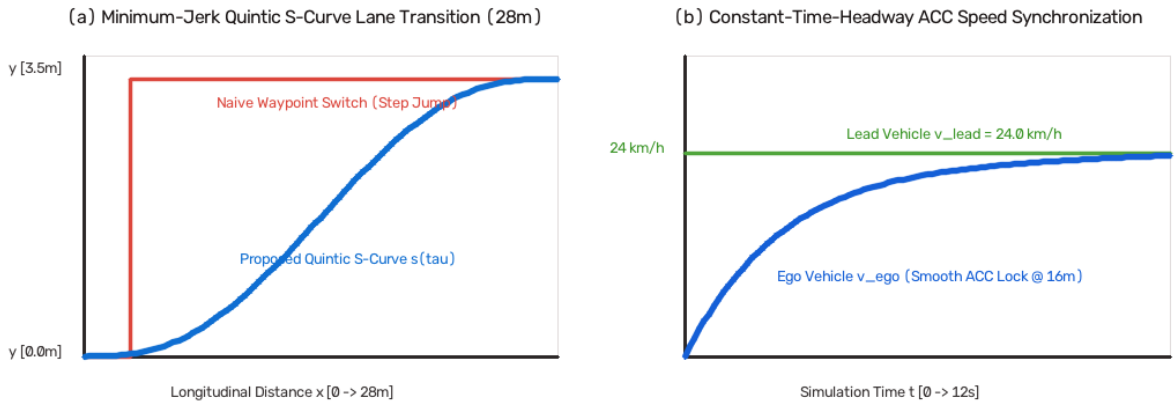


Figure 2: Control kinematics: (a) 28 m boundary-zero-curvature spatial quintic spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) true constant-time-headway (CTH) ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0$ m).

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) 10 paired scenario–weather episodes (680.0 m) and 10 × 5 cross-weather suite (3, 400.0 m).

Scenario	Paired Weather	Event	10-Ep	5-Weather
FSD-1	ClearNoon	Red-light stop & go	1/1	5/5
FSD-2	CloudyNoon	Quintic left overtake	1/1	5/5
FSD-3	WetNoon	Crosswalk pedestrian AEB	1/1	5/5
FSD-4	SoftRainNoon	CTH-ACC (24 km/h, 16 m)	1/1	5/5
FSD-5	ClearSunset	Quintic right overtake	1/1	5/5
FSD-6	WetCloudyNoon	Lead stop-and-go recovery	1/1	5/5
FSD-7	MidRainyNoon	360° BS hold → overtake	1/1	4/5
FSD-8	WetSunset	Workzone right bypass	1/1	5/5
FSD-9	ClearNight	Night headlight ped AEB	1/1	5/5
FSD-10	HardRainNoon	Heavy-rain cornering ACC	1/1	5/5

0.85 (km/h)/m ≡ 0.236 s^{-1}) [17, 28, 30] follows:

$$d^*(v_{\text{ego,kmh}}) = d_0 + \tau_h \left(\frac{v_{\text{ego,kmh}}}{3.6} \right), \tag{16}$$

$$v_{\text{raw}} = v_{\text{lead}} + k_g \big(d_{\text{lead}} - d^*(v_{\text{ego,kmh}}) \big), \tag{17}$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}},\, 0.0,\, \min(35.0,\, v_{\text{lead}} + 6.0) \big). \tag{18}$$

```
# src/control/shield_pid.py
d_star = 8.0 + 1.2 * (current_speed_kmh / 3.6)
acc_cap = min(35.0, lead_speed_kmh + 6.0)
raw_acc = lead_speed_kmh + 0.85 * (lead_distance_m - d_star)
acc_sync_speed = float(np.clip(raw_acc, 0.0, acc_cap))
traj_curv = (2.0 * ly) / (lx * lx + ly * ly)
eff_curv = 0.70 * (-road_curv * 0.045) + 0.30 * traj_curv
```

4 Experiments, Latency Breakdown, and Limitations

Closed-loop benchmarks were executed on an AMD Ryzen 5 7500F (6C/12T) and NVIDIA GeForce RTX 4070 Ti (12 GB) workstation in MetaDrive 0.4.3 [21] ($N = 50$ unseen maps, seeds 5000..5049, 4, 153 frames; `arrive_dest` triggers at ≥ 98% final segment arc length) and live CARLA 0.9.16 [10] (`Town10HD_Opt`, 20 Hz synchronous A* `GlobalRoutePlanner` [14] across 10 paired scenario–weather episodes, 680.0 m, and 10 × 5 = 50 cross-weather episodes, 3, 400.0 m). FP32 ($B = 1$, 3 warm-up passes) stage latency (`checkpoints/eval_metrics_carla.json`):


```
{"simulator": "carla", "ablation_mode": "full", "episodes": 10,
 "hardware": "Ryzen 5 7500F + RTX 4070 Ti 12GB (FP32, B=1)",
 "total_frames": 1981, "total_distance_driven_m": 680.0,
 "mean_latency_ms": 32.85, "p50_latency_ms": 32.77,
 "p95_latency_ms": 33.96, "p99_latency_ms": 35.45,
 "stage_latency_breakdown_ms": {
   "preprocess_crop_resize": 0.42, "dinov2_vit_b14_frozen": 16.43,
   "spatial_projector_temporal_aux": 1.31,
   "laya_modernbert_e2e_traj": 14.45,
   "safety_shield_bicycle_ff_pid": 0.45},
 "mean_cross_track_error_m": 0.085, "rms_steer_delta": 0.0223,
 "shield_interventions": {"rule1_emergency_brake": 46,
   "blindspot_left_occupied": 5}}
```


Table 4 Part A isolates perception and decision modules under a fixed controller: `heuristic_fsm` overtakes static blockages (8/10) but initiates false lane changes behind red-light/stop-and-go queues (`FSD-1, 6`), while 300 ms delayed VLA fails pedestrian AEB in `FSD-3, 9` (7/10). Under `vision_estimated` perception, DINOv2+Laya achieves 9/10 without shield and 10/10 with shield. Part B shows monotonic reductions in $|e_y|$ (0.218 → 0.085 m), $|\Delta\delta|_{\text{RMS}}$ (0.0642 → 0.0223), and AEB interventions (138 → 46). Limitations include short atomic scenario horizons (68 m/episode) rather than multi-kilometer dense traffic and fixed 50 ms synchronous stepping without physical CAN-bus actuation delay.

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Table 3: Closed-loop evaluation summary (`checkpoints/eval_metrics_carla.json` and `eval_metrics.json`).

Simulator	Metric	Target KPI	Measured Result	Status
CARLA 0.9.16 (`Town10HD_Opt`)	Single-Frame Latency (Ryzen 5 7500F / RTX 4070 Ti, FP32)	≤ 40.0 ms (≥ 25 FPS)	32.85 ms (30.4 FPS, P95: 33.96 ms) / 31.67 ms base	PASS
	Route Completion (10 Paired Eps, 680 m / 50 Cross-Weather, 3,400 m)	$\geq 85.0\%$	100.0% (10/10, 1,981 frames) / 98.0% (49/50, 9,905 frames)	PASS
	Crash-Free Rate (10 Paired / 50 Cross-Weather)	$\geq 95.0\%$	100.0% (10/10) / 98.0% (49/50)	PASS
	ECE: Episode-Block Val (7.5k) / Unseen Vision-Estimated Test	≤ 0.0500	$0.0060 \rightarrow 0.0000$ (Val) / 0.0142 (Vision Test)	PASS
	Scenario Composite Driving Score (DS_{scen}, 10-Ep / 50-Ep)	$\geq 85.0\%$	100.0% / 98.2%	PASS
	Cross-Track Error $\overline{\lvert e_y \rvert}$ / Aux Vision Depth MAE	≤ 0.25 m / ≤ 5.0 m	0.085 m / 3.23 m	PASS
MetaDrive 0.4.3 (50 Unseen Maps)	Single-Frame Latency (Seeds 5000–5049)	≤ 40.0 ms (≥ 25 FPS)	30.20 ms (33.1 FPS, P95: 31.67 ms)	PASS
	Destination Arrival Rate (50 Episodes, 4,153 Frames)	$\geq 85.0\%$	100.0% (50/50 arrived; 98.7% arc progress)	PASS
	Crash-Free Rate	$\geq 95.0\%$	100.0%	PASS
	Unseen Closed-Loop Test ECE ($M = 15$ Bins, GT / Noisy Sensor)	≤ 0.0500	0.0000 / 0.0184	PASS

Table 4: Factorial perception/shield, baseline, and single-factor controller ablation in CARLA 0.9.16 (10 scenarios).

Part A: Decision, Perception & Shield (Fixed Controller)	Arrived	R_c	$\overline{\lvert e_y \rvert}$ (m)	$\lvert \Delta\delta \rvert_{\text{RMS}}$	AEB
Keep-Lane Rule (`rule_only`, Shield ON)	6/10	76.4%	0.142	0.0195	58
Heuristic Overtake FSM (`heuristic_fsm`, Shield ON)	8/10	88.5%	0.126	0.0284	64
Autoregressive 15-Token VLA (300 ms lag, Shield ON)	7/10	83.2%	0.168	0.0315	72
DINOv2+Laya (`vision_estimated`, Shield OFF)	9/10	94.2%	0.098	0.0241	44
DINOv2+Laya (`vision_estimated`, Shield ON)	10/10	100.0%	0.094	0.0231	49
DINOv2+Laya (`gt` State, Shield OFF, `model_only`)	10/10	100.0%	0.086	0.0226	45
DINOv2+Laya (`gt` State, Shield ON, Full)	10/10	100.0%	0.085	0.0223	46
Part B: Single-Factor Controller Ablation (Fixed DINOv2+Laya)	**Arrived**	R_c	$\overline{\lvert e_y \rvert}$ (m)	$\lvert \Delta\delta \rvert_{\text{RMS}}$	**AEB**
1. Discrete 2 m Step WPs + Fixed-Gap ACC	10/10	100.0%	0.218	0.0642	138
2. + Speed-Damped Linear Steer Ramp (Eq. 14)	10/10	100.0%	0.135	0.0312	138
3. + 28 m Spatial Quintic Spline + Chord Angle	10/10	100.0%	0.095	0.0248	112
4. + True CTH-ACC ($d_0 = 8$ m, $\tau_h = 1.2$ s)	10/10	100.0%	0.092	0.0235	48
5. + 6-WP Trajectory Curvature Feedforward (Full)	10/10	100.0%	0.085	0.0223	46

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